

# 第二次作业说明

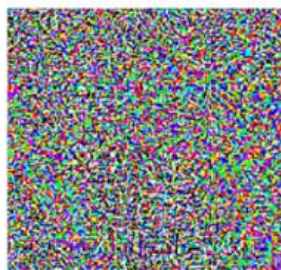
Cifar-10 数字世界  
无限制对抗攻击竞赛

近年来，人工智能技术在视觉识别领域有飞速的发展，但与此同时，人工智能模型的安全问题却不容乐观。

通过引入对抗样本，攻击者很容易就可以通过肉眼几乎观察不到的微小扰动，使模型分类失误。



原始样本  
机器：大熊猫  
人类：大熊猫



对抗噪音



对抗样本  
机器：长臂猿  
人类：大熊猫

本次比赛希望可以让参赛选手**了解和探索 Cifar10 上的对抗攻击场景**，通过组合对抗攻击方案，实地**体验不同对抗攻击算法**特点。

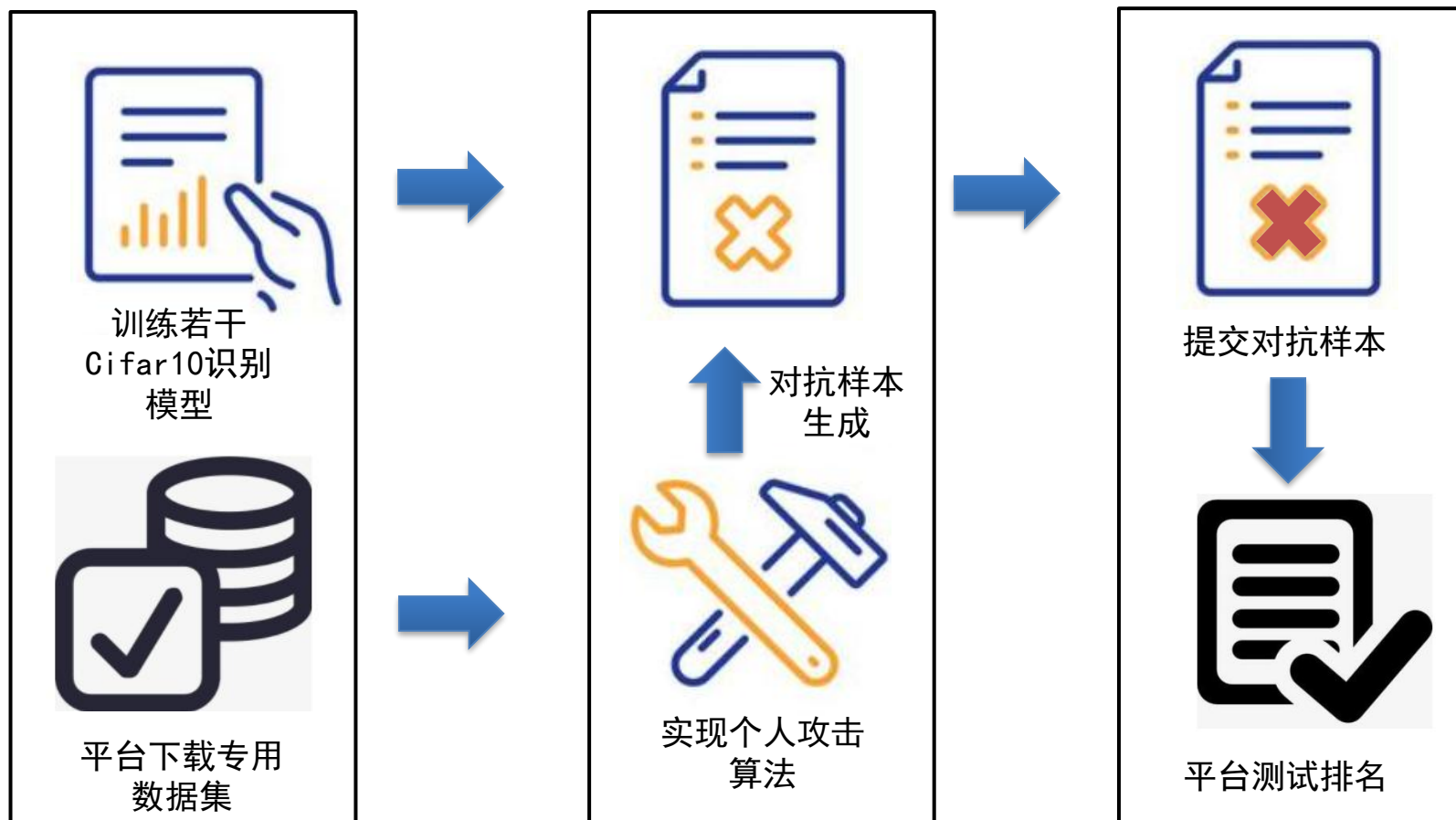
# 比赛流程：总览

第一步：参赛者可通过<https://challenge.aisafety.org.cn/#/competitionDetail?id=26>

，注册并下载本赛事提供的专用数据集

第二步：参赛者本地实现图像攻击算法，并对专有数据集做攻击

第三步：参赛者平台中提交个人攻击后数据集，数据集格式与原数据集格式相同



# 比赛流程：总览

## 一 基本规则

- 用户通过平台注册、组队，每支队伍最多**3**名成员（按deepfake大作业分组）。
- 比赛开始时放出训练集和测试集，按照平台要求的格式上传测试集的推理结果，平台会综合计算最终得分。
- 每队每天最多上传**3**次结果。

## □ 时间安排

- 组队&熟悉平台和赛题：5.15-5.17
- 提交开始：5.17
- 答辩汇报：6.26
- 比赛结束&评分：6.26



# 比赛流程：文件提交

## ➤ 数据集说明

- 本赛事采用 Cifar-10 数据集，我们预先筛选了500张图像，所有图像均为  $32 * 32$  尺寸。
- 图像命名方式为 X.jpg，其中 X 为  $[0, 500)$  范围内的整型数字。
- 比赛提供了标签文件 label.txt，该文件共 500 行，其中每一行代表：  
图像名称      图像类别

## ➤ 提供的数据集目录树

```
-- images
    |-- 0.jpg
    |-- 1.jpg
    |-- ...
    |-- 499.jpg
-- label.txt
```

## ➤ 提交结果

- 参赛者最终需要提交一个**zip压缩包**，压缩包目录树应与上方目录树相同
- 参赛者不应修改label.txt文件
- 参赛者不应修改jpg名称，即仍命名为0~499.jpg。

# 评分标准：总览

本次竞赛共分为**客观分**和**主观分**两部分。其中总体成绩中，主观分占比50%，客观分占比50%。

客观分通过平台自动评测，主要测试以下几方面：

1. 攻击性：即参赛者所提交样本的攻击能力
2. 修改力度：即参赛者对原始样本的变更程度

客观分部分评分标准为：**两种指标计算结果连乘结果**

主观分通过人工筛查，检查违规操作。主要测试以下几方面：

1. 图像是否合乎规范
2. 所提交图像的质量

主观分部分评分标准为：**两种指标计算结果相乘结果**

# 评分标准：攻击成功率ASR

本次竞赛采用攻击成功率ASR(Attack Success Rate)，衡量参赛者提交样本的攻击性。平台**不公开**的实现有若干判别模型，且针对初始的500张样本，判别模型预测成功率均为**100%**。

用户执行对抗攻击后，若使模型出现误分类，说明攻击成功，最终攻击成功率采用如下公式计算：

$$Score_{ASR} = \frac{\text{攻击成功样本数}}{\text{总体样本数}}$$

更形式化的：

$$Score_{ASR} = \frac{\|x' | F(x') \neq y\|}{n}$$

其中 n 表示样本数，F(x)表示模型预测结果，y 表示原始标签

# 评分标准：图像结构相似度SSIM

本次竞赛采用图像结构相似度SSIM(Structural Similarity)，衡量参赛者提交样本，是否很大幅度的偏离了原始样本。

其中具体的SSIM公式为：

$$\mu_x = \frac{1}{N} \sum_{i=1}^N x_i$$
$$\sigma_x = \left( \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_x)^2 \right)^{\frac{1}{2}}$$
$$Score_{SSIM}(x, x') = \frac{(2\mu_x\mu_{x'} + C_1)(2\sigma_{xx'} + C_2)}{(\mu_x^2 + \mu_{x'}^2 + C_1)(\sigma_x^2 + \sigma_{x'}^2 + C_2)}$$

具体的，在本次竞赛中，测试的是攻击成功样本的平均图像结构相似度。更多的结构相似度理论说明，可以参看：

[《Image Quality Assessment : From Error Visibility to Structural Similarity》](#)

总体的客观部分打分规则为

$$Score_M = 100 * Score_{ASR} * Score_{SSIM}$$

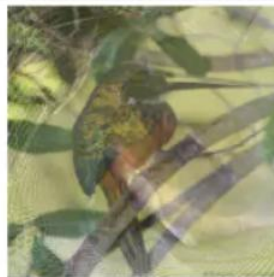
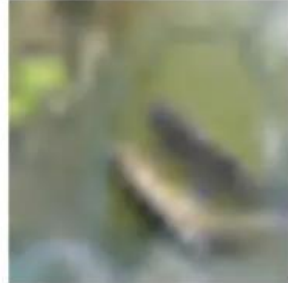


# 评分标准：合规图像筛选

本次竞赛除平台自动判分外，最终还将组织人工评测，筛选不合规图像。并按照合规图像比例给予该部分分值。

$$Score_{valid} = \frac{\text{合格样本数}}{\text{总体样本数}}$$

具体的，不合规示例如下：

|          |                                                                                         |                                                                                             |                                                                                            |                                                                                     |                                                                                              |                                                                                               |                                                                                              |
|----------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 语义一致     | <br>原图 | -                                                                                           | 混入其他语义                                                                                     |  | <br>MixUp | <br>CutMix |                                                                                              |
| 直接用其他图替换 |       | <br>其他类图像 | <br>同类图像 | 过度的图像变换                                                                             |         | <br>过度模糊  | <br>风格迁移 |

# 评分标准：图像质量评分

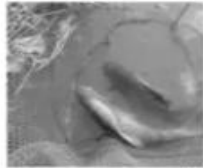
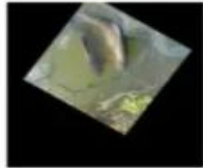
每张图像，按照肉眼辨识度评分，最高分为5分，最低分为1分。

$$Score_{Quality} = \frac{\sum \text{样本分数}}{5 * \text{样本数}}$$

具体的，打分规则如下：

| $Score_q$ | 描述                      | 范例图像1                                                                                              | 范例图像2 |
|-----------|-------------------------|----------------------------------------------------------------------------------------------------|-------|
| 5         | 图像质量最高，原图和攻击后的图从视觉上基本一致 | <br>原图          | —     |
| 4         | 图像质量较好，可以清晰辨认目标         | <br>Elastic 变换 | —     |

# 评分标准：图像质量评分

|   |                       |                                                                                                |                                                                                               |
|---|-----------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 3 | 图像质量一般                | <br>灰度化     | <br>JPEG压缩 |
| 2 | 图像质量较差，有视觉上的干扰，不能一眼辨认 | <br>仿射变换    | <br>透视变换   |
| 1 | 图像质量极差，只能勉强辨认目标       | <br>较大的高斯噪声 | <br>低对比度   |

总体的主观部分打分规则为

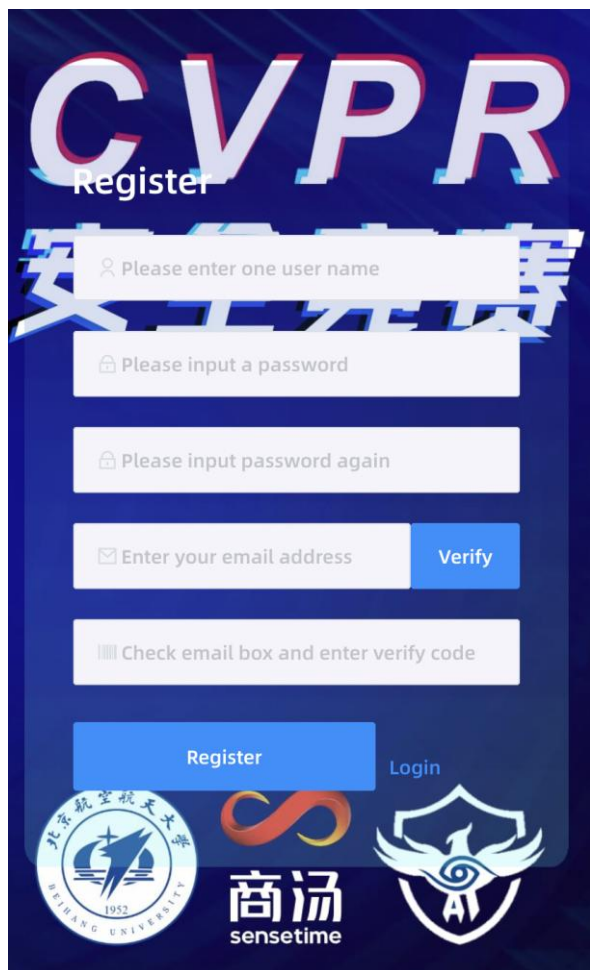
$$Score_H = 100 * Score_{valid} * Score_{quality}$$

总体的计分规则为

$$Score = 0.5 * Score_M + 0.5 * Score_H$$

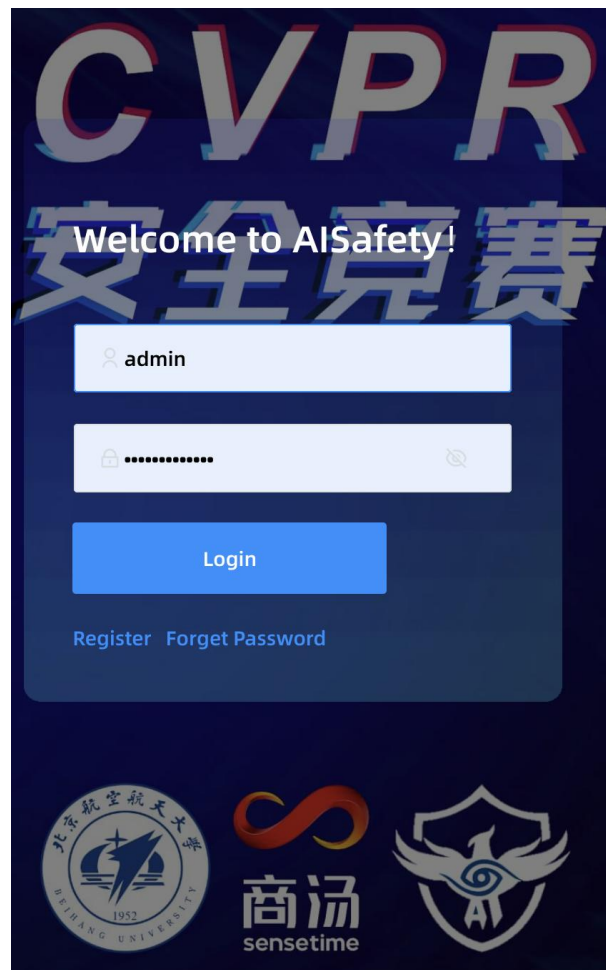
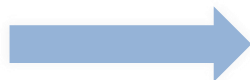
# AISafety比赛平台

- 平台地址: <https://challenge.aisafety.org.cn/#/competitionDetail?id=26>



The registration form is titled "CVPR" and "AISafety". It includes a "Register" button at the top left. The form fields are: "Please enter one user name", "Please input a password", "Please input password again", "Enter your email address" (with a "Verify" button), and "Check email box and enter verify code". At the bottom, there are logos for Beijing University of Aeronautics and Astronautics (1952), Sensetime, and AIS.

注册



The login form is titled "CVPR" and "AISafety". It includes a "Welcome to AISafety!" message. The form fields are: "admin" (username) and "password" (masked with dots). There is a "Login" button and links for "Register" and "Forget Password". At the bottom, there are logos for Beijing University of Aeronautics and Astronautics (1952), Sensetime, and AIS.

登录

# AI Safety 比赛平台

## — 比赛详情页

 AISafety

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菜单

查看不同比赛内容介绍

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Phase II

Track I: Classification Task Defense

Robust Models towards Open-world Classification

160 Teams · Phase I up to 2022-04-21 20:00:00 · Phase II up to 2022-05-21 20:00:00

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Introduction

Deep learning models are vulnerable to noises, e.g., adversarial attacks, which pose threats to deep learning-based applications in real-world scenarios. A large number of defense methods have been proposed to mitigate the threats, but most of them can be bypassed, and their robustness on large-scale datasets is still far from satisfactory.

The purpose of this track is to accelerate the research on building robust models against noises and novel defense algorithms. Participants are encouraged to develop defense methods that improve model robustness against diverse noises on large-scale datasets.

You can refer to the model robustness benchmark at [RobustART](#).

Rules of the competition

All teams must register between 2022-3-11 20:00:00 - 2022-4-10 18:00:00. Each team can have no more than 5 members. Note



# AI Safety 比赛平台

## — 数据集简介及下载

 AISafety

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[Competitions](#)  
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Phase II

### Track I: Classification Task Defense

#### Robust Models towards Open-world Classification

160 Teams · Phase I up to 2022-04-21 20:00:00 · Phase II up to 2022-05-21 20:00:00

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Data Description

## Datasets

会从此页面释放出数据集下载链接

Phase I | training set: [Click to download \(Google Drive\)](#) [Click to download \(Baidu Drive, code: pp1f\)](#)

Phase I | test set1: [Click to download \(Google Drive\)](#) [Click to download \(Baidu Drive, code: 13gp\)](#)

Phase I | test set2: [Click to download \(Google Drive\)](#) [Click to download \(Baidu Drive\)](#)

Phase II training set: [Click to download \(Google Drive\)](#) [Click to download \(Baidu Drive\)](#)

**In Phase I**, we will release the training set and test set. The training set contains 22986 clean images with 20 classes (labeled as [0,19]). The test set contains 20000 perturbed images(e.g., adversarial examples, image corruptions, etc.) with the same 20 classes. **Note that we do not constrain the type and degree of perturbations. One image may be added multiple noises.** You are encouraged to generate various perturbed samples based on the training set to train your robust classifier. **To avoid overfitting to the given test samples, we will release another test set with 5540 new perturbed images one day before the end of Phase I.** Finally we will calculate the eventual Phase I score based on 50% score on the first test set and 50% score on the second test set

# AI Safety 比赛平台

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## Track I: Classification Task Defense

### Robust Models towards Open-world Classification

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New Topic

### Discussion

Type in ...

Search

All

Owned

| # | Author        | Title                                                    | Comments |
|---|---------------|----------------------------------------------------------|----------|
| 1 | paulgavrikov  | <a href="#">Private questions</a>                        | 1        |
| 2 | dmitrii.korzh | <a href="#">Submission error</a>                         | 1        |
| 3 | cold          | <a href="#">Take a break and enjoy some funny memes!</a> | 0        |

可在此处发表帖子与他人交流或问题反馈

# AI Safety 比赛平台

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#### Leaderboard

Type in ...

Search

Phase I

Phase II

| # | Team         | Members                                                                                                                                                                                                                                                                                                                                                                                                                              | Score (set 1) | Score (set 2) | Final score ⓘ | Last                |
|---|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------------|
| 1 | team_AIPower |                                                                                                                                                                             | 96.92         | 97.06         | 96.99         | 2022-04-18 12:33:01 |
| 2 | RobustX      |      | 96.66         | 96.72         | 96.69         | 2022-04-18 10:55:36 |
| 3 | HotEye       |                                                                                                                                                                                                                                                                                                                                                   | 96.23         | 96.37         | 96.3          | 2022-04-17 21:35:23 |



# AI Safety 比赛平台

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Manage Team

Team Name

team\_admin

Save Team Name

可以修改自己的队伍名称

Team Members (1 of 5 maximum)

1



admin (you)

Leader

Agreed

Invite Member

Invite another user to join with your.

Type in other user's username.

Send

可在此邀请他人进入自己的队伍

Pending Team Invitation

如果有收到他人的邀请，这里会显示是否接受

No Data

# AI Safety 比赛平台

## — 提交结果



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Refresh

|    | Submit Time         | Submitter | Status   | Score Type | Score | Submission Data             |
|----|---------------------|-----------|----------|------------|-------|-----------------------------|
| 1  | 2022-04-28 11:51:53 | admin     | FAILED ⓘ | Phase II   | -     | <a href="#">View Result</a> |
| 2  | 2022-04-28 11:37:04 | admin     | RUNNING  | Phase II   | -     | <a href="#">View Result</a> |
| 3  | 2022-04-27 11:57:16 | admin     | FAILED ⓘ | Phase II   | -     | <a href="#">View Result</a> |
| 4  | 2022-04-27 00:31:23 | admin     | FAILED ⓘ | Phase II   | -     | <a href="#">View Result</a> |
| 5  | 2022-04-26 10:04:46 | admin     | ERROR ⓘ  | Phase II   | -     | <a href="#">View Result</a> |
| 6  | 2022-04-23 17:51:40 | admin     | FAILED ⓘ | Phase II   | -     | <a href="#">View Result</a> |
| 7  | 2022-04-23 17:39:24 | admin     | ERROR ⓘ  | Phase II   | -     | <a href="#">View Result</a> |
| 8  | 2022-04-23 13:03:51 | admin     | SUCCESS  | Phase II   | 52.96 | <a href="#">View Result</a> |
| 9  | 2022-04-23 11:51:07 | admin     | FAILED ⓘ | Phase II   | -     | <a href="#">View Result</a> |
| 10 | 2022-04-22 23:11:20 | admin     | FAILED ⓘ | Phase II   | -     | <a href="#">View Result</a> |

Total 14

10/page

<

1

2

>

Go to

1

▲

# AI Safety 比赛平台

## — 上传json文件

The screenshot displays the AISafety Competition Detail page. A modal dialog titled "Submit Predictions" is open in the center. The dialog contains a blue instruction box: "Please upload files according to the rules. Support .onnx, .zip file. File size <= 160MB". Below this is a dashed box with a cloud upload icon and the text "Drop file here or [click to upload](#)". At the bottom right of the modal are "Cancel" and "Upload" buttons. The background page shows the competition details for "Phase II" with a table of submissions.

**Submit Predictions**

Please upload files according to the rules. Support .onnx, .zip file. File size <= 160MB

Drop file here or [click to upload](#)

Cancel Upload

|   | Score | Submission Data             |
|---|-------|-----------------------------|
| 1 | -     | <a href="#">View Result</a> |
| 2 | -     | <a href="#">View Result</a> |
| 3 | -     | <a href="#">View Result</a> |
| 4 | -     | <a href="#">View Result</a> |
| 5 | -     | <a href="#">View Result</a> |
| 6 | -     | <a href="#">View Result</a> |
| 7 | -     | <a href="#">View Result</a> |

# AI Safety 比赛平台

## □ 比赛流程

- “注册” → “登录” → 选择菜单 “Competitons” , 点击对应的比赛 → 点击右侧 “Join Competition” 先申请加入该比赛。
- 加入比赛成功后即可在 “Data” 标签页面下载相应的数据集, 训练后将测试集推理结果按照比赛介绍页的json格式, 通过点击 “Submit Predictions” 按钮上传至服务器端。
- 当服务器端评判结束后, 即可在 “My Submission” 页查看到相应提交记录和分数。

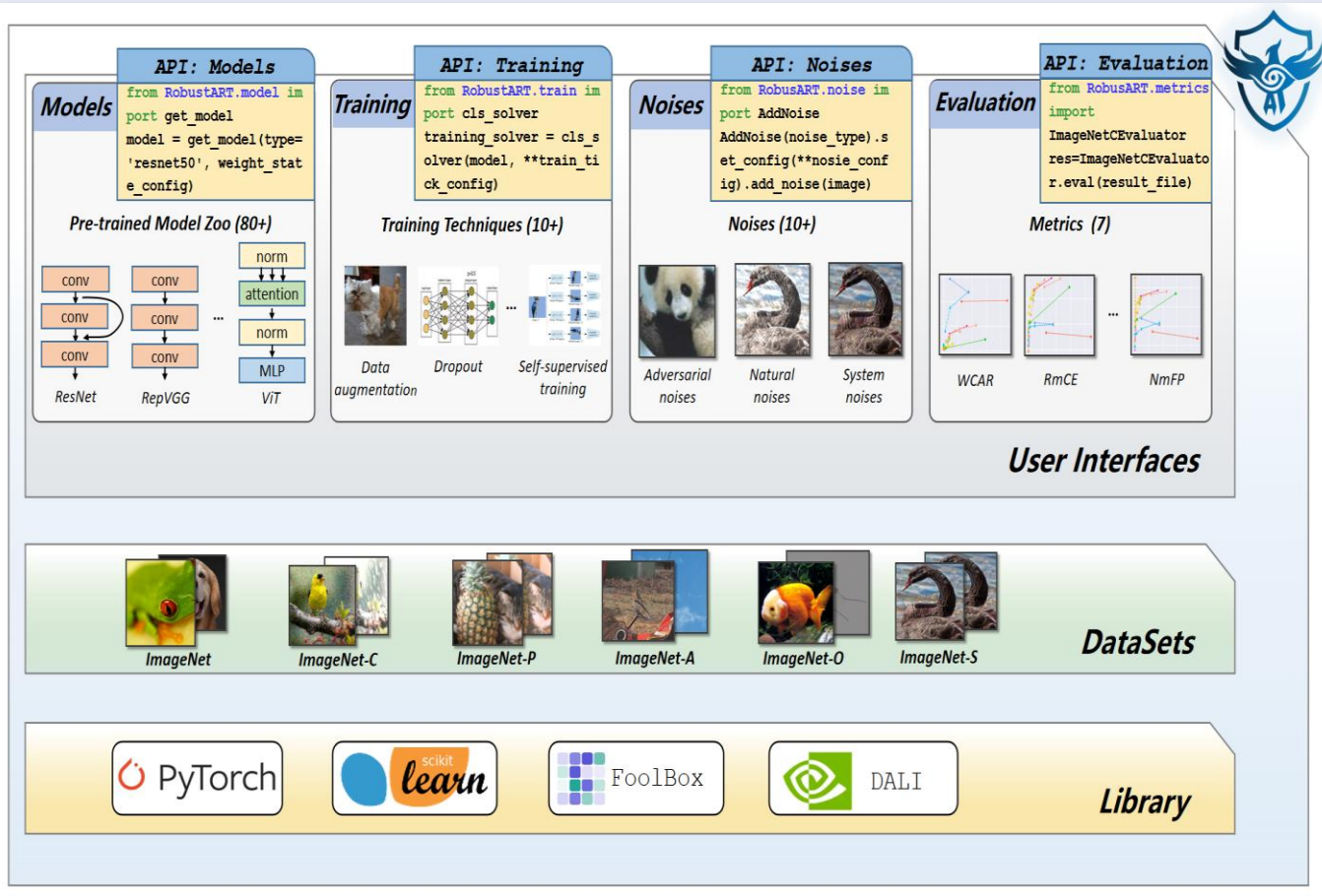
## □ 其他页面

- Overview标签页可以查看比赛详细介绍, 包括比赛描述、时间线、代码要求、评价指标等。
- Data标签页可以查看释放出的训练集、测试集及说明。
- Discussion标签页可以发表论坛帖子和查看他人帖子。
- LeaderBoard标签页可以查看当前实时排行榜。
- Team标签页可以修改队名、邀请他人入队、接受他人邀请等。
- My Submission标签页可以看到队伍成员的提交记录及状态等。

# RobustART鲁棒评测基准

## RobustART生态框架

- 四大核心模块
  - Models模块
  - Training模块
  - Noises模块
  - Evaluation模块
- 用户可自由扩展
  - 提供扩展API
  - 可自定义模型等
- 框架和平台完全开源, 且提供80余种预训练模型供下游任务

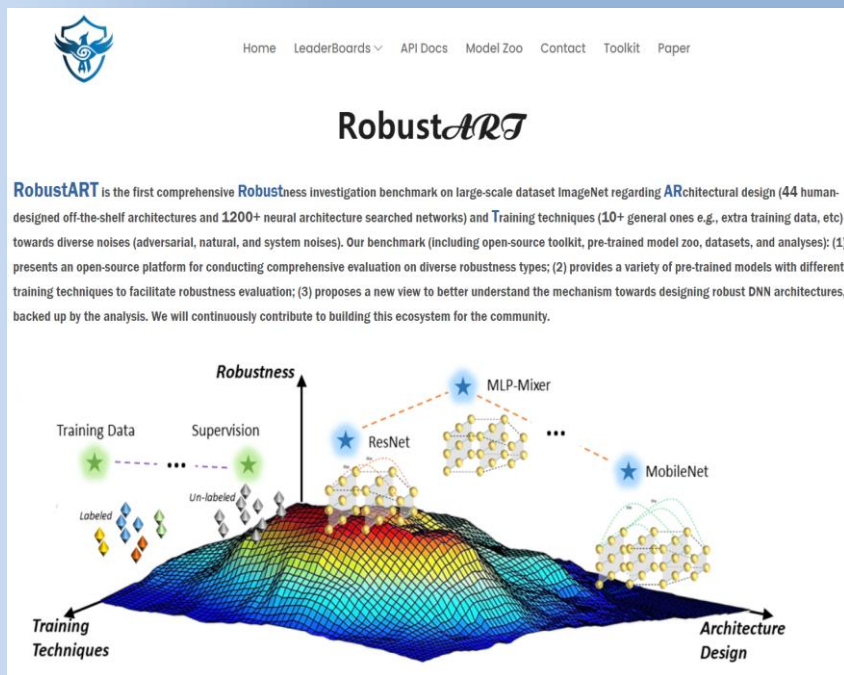


\* RobustART取自Robust benchmark on ARchitecture design and Training techniques.

# RobustART鲁棒评测基准

## RobustART 开放平台

<http://robust.art/>



RobustART 平台主页

| Show 10 entries <span>Search:</span> |                    |           |                |          |           |                               |                               |                                 |
|--------------------------------------|--------------------|-----------|----------------|----------|-----------|-------------------------------|-------------------------------|---------------------------------|
| Model Architecture                   | Training Technique | Params(M) | Conv Params(M) | Flops(G) | Clean Acc | FGSM Adv ( $\epsilon=8/255$ ) | FGSM Adv ( $\epsilon=2/255$ ) | FGSM Adv ( $\epsilon=0.5/255$ ) |
| DeiT-B                               | Naive              | 86.568    | 0.591          | 17.6     | 66.412    | 12.488                        | 27.964                        | 51.535                          |
| DeiT-S                               | Naive              | 22.051    | 0.295          | 4.6      | 65.604    | 8.828                         | 20.684                        | 43.482                          |
| DeiT-Ti                              | Naive              | 5.717     | 0.148          | 1.193    | 65.474    | 4.319                         | 8.876                         | 30.103                          |
| DenseNet-121                         | Naive              | 7.979     | 6.87           | 2.833    | 74.858    | 16.815                        | 20.016                        | 40.831                          |
| DenseNet-169                         | Naive              | 14.149    | 12.326         | 3.358    | 75.938    | 18.528                        | 22.199                        | 43.161                          |
| DenseNet-201                         | Naive              | 20.014    | 17.864         | 4.283    | 76.488    | 22.016                        | 25.998                        | 46.009                          |
| EfficientNet-B0                      | Naive              | 5.289     | 3.966          | 0.385    | 69.262    | 1.146                         | 2.85                          | 34.258                          |
| EfficientNet-B0                      | Dropout            | 5.289     | 3.966          | 0.385    | 68.39     | 0.912                         | 2.146                         | 30.875                          |
| EfficientNet-B1                      | Naive              | 7.794     | 6.451          | 0.685    | 70.254    | 1.372                         | 2.863                         | 32.43                           |
| EfficientNet-B2                      | Naive              | 9.11      | 7.633          | 0.992    | 71.282    | 1.248                         | 2.744                         | 32.271                          |

Showing 1 to 10 of 88 entries

Previous 1 2 3 4 5 ... 9 Next

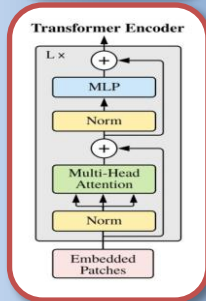
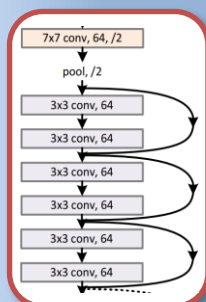
Leaderboard

# RobustART鲁棒评测基准

## 核 心 思 想

### 模型架构和训练技巧的解耦

不同网络架构的模型  
使用同样的训练  
技巧训练

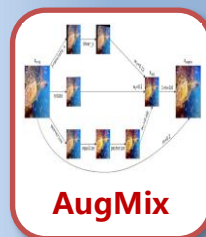


+

- Adam
- Opt
- AutoAug
- Dropout
- Normaliz e
- ECosine
- .....

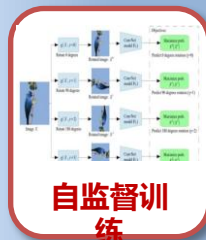
全面的鲁棒性评估

同种网络架构分别  
使用不同训练技巧  
训练



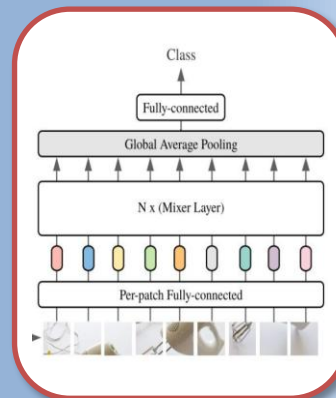
AugMix

+



自监督训练

.....





# RobustART鲁棒评测基准

## 模型架构仓库

**44** 种不同经典网络架构，覆盖三大主流模型架构：CNN、Transformer、MLP-Mixer；**1200** 余种NAS网络架构

Table 2: Architecture repository.

| Category    | Arch. Family      | Archs                                                                                      |
|-------------|-------------------|--------------------------------------------------------------------------------------------|
| CNN         | ResNet [28]       | ResNet-18, ResNet-34, ResNet-50, ResNet-101, ResNet-152                                    |
|             | ResNeXt [56]      | ResNeXt-50-32x4d, ResNeXt-101-32x8d                                                        |
|             | WideResNet [57]   | WRN-50-2, WRN-101-2                                                                        |
|             | DenseNet [29]     | DenseNet-121, DenseNet-169, DenseNet-201                                                   |
|             | ShuffleNetV2 [58] | ShuffleNetV2-x0.5, ShuffleNetV2-x1.0, ShuffleNetV2-x1.5, ShuffleNetV2-x2.0                 |
|             | MobileNetV2 [59]  | MobileNetV2-x0.5, MobileNetV2-x0.75, MobileNetV2-x1.0, MobileNetV2-x1.4                    |
|             | RepVGG [60]       | RepVGG-A0, RepVGG-B3                                                                       |
|             | RegNetX [61]      | RegNetX-400M, RegNetX-800M, RegNetX-1600M, RegNetX-3200M, RegNetX-6400M                    |
|             | EfficientNet [30] | EfficientNet-B0, EfficientNet-B1, EfficientNet-B2, EfficientNet-B3, EfficientNet-B4        |
| Transformer | MobileNetV3 [62]  | MobileNetV3-x0.35, MobileNetV3-x0.5, MobileNetV3-x0.75, MobileNetV3-x1.0, MobileNetV3-x1.4 |
|             | ViT [31]          | ViT-B/16, ViT-B/32                                                                         |
|             | DeiT [32]         | DeiT-Ti, DeiT-S, DeiT-B                                                                    |
| MLP         | MLP-Mixer [33]    | Mixer-B/16, Mixer-L/16                                                                     |

# RobustART鲁棒评测基准

## 训练技巧仓库

**11** 种不同的训练技巧，划分为五大类：训练监督、权重调整、正则化、训练数据、优化器

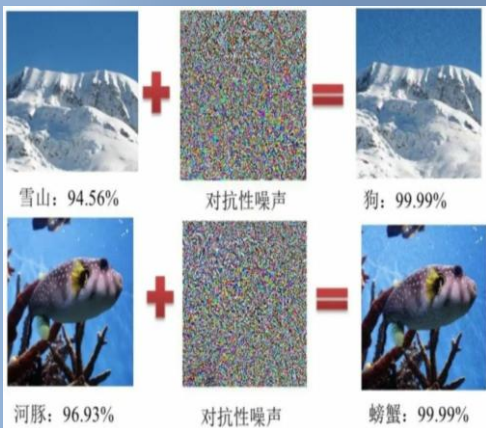
Table 3: General training techniques.

| Category             | Factors                                                                               |
|----------------------|---------------------------------------------------------------------------------------|
| Training Supervision | Knowledge Distillation [63], Self-supervised Training [64]                            |
| Weight Adjustment    | Weight Averaging [65], Weight Re-parameterization [60]                                |
| Regularization       | Label Smoothing [66], Dropout [67]                                                    |
| Training Data        | Data Augmentation [68] [69], Large-scale Pre-training [70], Adversarial Training [22] |
| Optimizer            | SGD [71], AdamW [72]                                                                  |

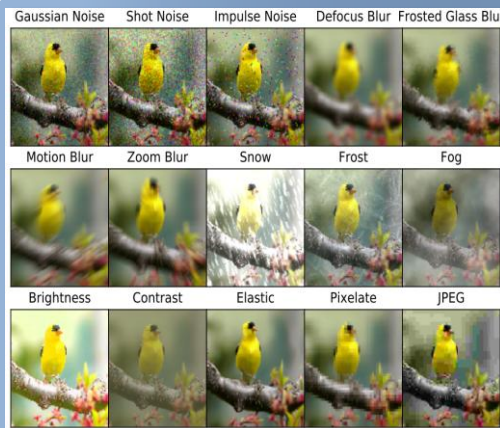
# RobustART鲁棒评测基准

## 多种鲁棒性评估噪音

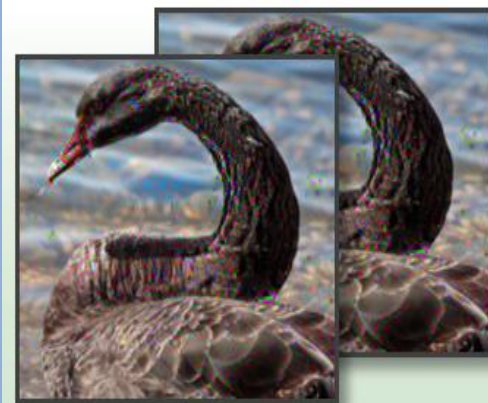
提供全面的鲁棒性评估，覆盖**3**大类型噪音：包含**8**种对抗攻击方法、**4**种自然噪音数据集、**1**种系统噪音数据集



对抗噪音 (如 AutoAttack, PGD)



自然噪音 (如 ImageNet-C)



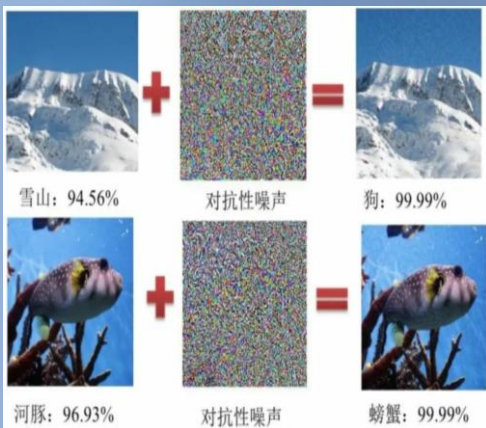
系统噪音 (如 ImageNet-S)

基于大规模数据集 ImageNet

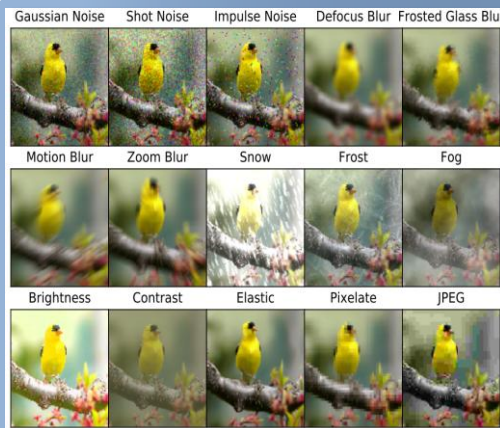
# RobustART鲁棒评测基准

## 多种鲁棒性评估噪音

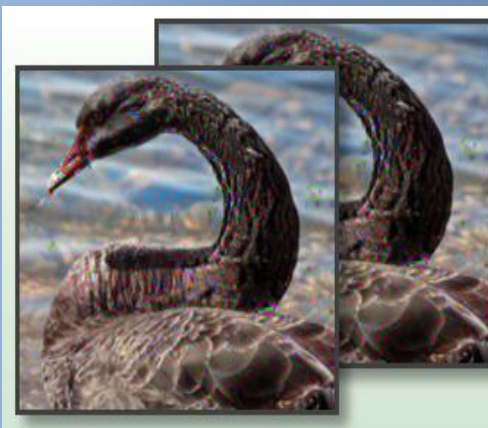
提供全面的鲁棒性评估，覆盖**3**大类型噪音：包含**8**种对抗攻击方法、**4**种自然噪音数据集、**1**种系统噪音数据集



对抗噪音 (如 AutoAttack, PGD)



自然噪音 (如 ImageNet-C)



系统噪音 (如 ImageNet-S)

基于大规模数据集 ImageNet

**谢谢聆听！**